Abhishek Mhamane

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Education

•Indian Institute of Technology, Kanpur *2022-Present*
MS-Research in Geoinformatics **CPI: 9.11**
Relevant Coursework: Physical Geodesy (A*), Environmental Geodesy (A)
Adjustment Computation, Introduction to GNSS

•Visvesvaraya National Institute of Technology(VNIT), Nagpur *2018-22*
B.Tech in Civil Engineering **CGPA: 8.27**
Relevant Coursework: AI for Engineers, Computer Programming,
Numerical Methods, Remote Sensing and GIS, Surveying, Strength of Materials

•Padmashri Shankar Bapu Apegoankar(PSBA) English Scool, Aurangabad *2017-18*
Class XII, CBSE **Percentage: 93.6 %**

•Padmashri Shankar Bapu Apegoankar(PSBA) English Scool, Aurangabad *2015-16*
Class X, CBSE **CGPA: 10.0**

Research Experience

•SOCE Summer Intern *May'23 - July'23*
Indian Institute of Technology, Kanpur Kanpur

- Under Prof. Balaji Devaraju and PhD Scholar Ibaad Anwar of IIT Kanpur
- Automating the data transfer pipeline using Google Drive API
- Python module with wrapper functions

•Sumer Research Intern *May'21 - July'21*
Visvesvaraya National Institue of Technology, Nagpur Virtual

- under the guidance of prof. Ashwini Mirajkar
- Developed Spatial Multi-Criteria Analysis using AHP technique in GIS
- Implemented Dijkstra's Algorithm to find LCP on a isotropic surface
- Proposed path reduced length by 17% when compared to government planned route

Projects

•Regional Geoid Modelling using GRAV-D Airborne gravimetric data *Sept'22-Dec'22*
Course Project: CE678A-Physical Geodesy

- Adopted Remove-Compute-Restore approach for Geoid determination
- Computed gravity anomalies, applying topographic correction and Stokes Integral using FFT
- Developed OCTAVE toolbox for processing GRAV-D data

•Topographic Mapping, GPS Survey and Control Network *Dec'22*
Course Project: CE673A-Survey Camp

- Coordinated a team of 4 on a week long mapping exercise
- Instruments - Total Station, GNSS receivers, Hand-Held GPS, Auto level
- Appreciating the cartography

•Structural Analysis Interactive Web App (Teaching Aid) *May'22 - July'22*
Visvesvaraya National Institute of Technology, Nagpur GitHub Repository

- Developed a Web App teaching aid for Structural Analysis course
- Simple and Interactive app with Beam Analysis, Plotting and Visualization features
- **Tools & technologies**: Python, Web development, Science Communication

•ANN based Soil Classification from first principles *March'22 - May'22*
Course Project MEL457-AI for Engineers [VNIT] GitHub Repository

- Utilised LandSat data for classification
- Developed ANN from scratch & visualized "Black-Box" nature of ANN
- Published the results as Web App
- **Tools & Technologies**: Python, Sci-Kit Learn, StreamLit

•Mapathon Organized by IIT Bombay and ISRO *Dec 2020*

Change Detection Analysis using Open Source Remote Sensing data

- Carried out change detection analysis of Nagpur city b/w 2008 to 2018
- Utilized LISS-3 satellite data, carried out DOSM atmospheric correction
- NDVI, NDWI and other indices used for change detection in QGIS
- **Skills**: QGIS, Geospatial Analysis

Open Source Contributions

•PySHBundle *May'22 - present*

A Python implementation of MATLAB codes SHbundle initiative by IISc Bangalore GitHub Repository

- Under the guidance of Prof. Bramha Dutt Vishwakarma
- Contributed new features as well as documenting the project
- Debugged code
- Currently in process of publishing the research paper

Publications

Least Cost Path Pipeline Routing Using Spatial Multi-criteria Analysis for Vidarbha Region: A Case Study

Mhamane, A., Mirajkar, A.B. (2023). Least Cost Path Pipeline Routing Using Spatial Multi-criteria Analysis for Vidarbha Region: A Case Study. In: Timbadiya, P.V., Patel, P.L., Singh, V.P., Mirajkar, A.B. (eds) Geospatial and Soft Computing Techniques. HYDRO 2021. Lecture Notes in Civil Engineering, vol 339. Springer, Singapore. https://doi.org/10.1007/978-981-99-1901-7_30

Achievements

•Institute Fellowship *2022-present*

MS-Research, IIT Kanpur

•All India Rank (AIR) - 22 | GATE Geomatics Engineering *Feb'22*

Graduate Apptitude Test in Engineering (GATE) Entrance Exam for IITs and IISc

•Prime Minister's Scholarship Scheme *2018-22*

Ministry of Defence, Government of India

Scholarship Awarded for the 4 years of undergraduate education at VNIT Nagpur

•National Bal Shree Honour *2014*

National Bal Bhavan, Ministry of Human Resource Development, Government of India

Selected for a week long camp at National Bal Bhavan, New Delhi
after series of competitions at district level (Porbandar) and Zonal Level (Goa Bal Bhavan).

Certification

•GRACE HackWeek-2 *Sept'22*

Online School Organized by IIT Kanpur on GRACE Data Processing

•Python3 Programming Specialization *May'22*

University of Michigan | Coursera

•Web-Developmnet and Coding Specialization *May'22*

University of Michigan | Coursera

Technical Skills

Programming Languages: Python, MATLAB/OCTAVE, C, JavaScript
Tools/Softwares : QGIS, ArcGIS, Git, LaTeX, Linux
Instruments : Total Station, Auto Level, GNSS receivers, Relative Gravimeters

Language Fluency

English: Full professional fluency
Marathi and Hindi: Native speaker
Japanese: Begineer learner